

MATH 1342 – Elementary Statistics – Week 4: One Quantitative Variable: Shape, Center, Spread and Percentiles

Sep 14 – 16 | 2.2 One Quantitative Variable: Shape and Center • 2.3 One Quantitative Variable: Percentiles | Fall 2026 • Section 019 • MW 7:30–8:50 PM • RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions.	1. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CO1) 2. Compute the mean and median of a dataset and explain how skewness and outliers affect each (resistance). (CO2) 3. Interpret the standard deviation as the typical distance from the mean and apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CO2, CO4) 4. Compute the sample standard deviation of a small dataset by hand. (CO2) 5. Compute and interpret z-scores to compare values measured on different scales or in different groups. (CO2, CO4) 6. Find the five-number summary, range, and IQR; apply the 1.5×IQR rule to identify outliers; and construct or interpret a boxplot. (CO1, CO2)	e-Text: 2.2 One Quantitative Variable	MO1, MO2, MO3	Participation: Quantitative Variables – shape, mean vs. median; estimate the mean and SD of a dotplot without calculating (95% rule)	MO1, MO2, MO3	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 20, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 2.2 One Quantitative Variable	MO1, MO2, MO3				
		Slides: 2.2 One Quantitative Variable	MO1, MO2, MO3	Participation: One Quantitative Variable (Kahoot!) – Wed: review of 2.2	MO1, MO2, MO3	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)	MO4, MO5, MO6
		e-Text: 2.3 One Quantitative Variable: Using Percentiles	MO4, MO5, MO6	Participation: Using Percentiles – Your Turn: comparing heights with z-scores; five-number summary	MO4, MO5, MO6		
		Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles	MO4, MO5, MO6				
		Slides: 2.3 One Quantitative Variable: Using Percentiles	MO4, MO5, MO6			Take-Home Quiz 2 (due Mon Sep 21, 11:59 PM)	All